

ASU Macro Qual Module 1

Optimization and Dynamic Programming

Exhaustive, Exam-Facing Lecture Notes Packet

Built from the ASU Macro Manifest, PS1, and uploaded 2016–2024 qualifying exams

May 2026

Purpose. This packet is the mathematical foundation module for the ASU Macro Theory qualifying exam. It is designed to train the actions that repeatedly earn points: formulate a problem recursively, identify the state and control, write the feasible correspondence, derive first-order and envelope conditions, combine them into Euler equations, prove basic dynamic-programming properties, and execute closed-form guess-and-verify derivations under time pressure.

Target weakness. Recursive formulation and closed-form derivations. Every canonical example is written in the sequence that should appear on an exam: environment, timing, state, control, Bellman equation, FOC/KKT, envelope, Euler or policy rule, closed form, intuition, trap.

Expected study use. Two 3-hour blocks for the core theory and canonical models; one 3-hour block for closed-form practice; one 2-hour block for exercises and answer sketches. A student who is weak on recursion should start with Sections [6](#), [7](#), [14](#), and then solve the exercises before reading answer sketches.

Contents

1	Source inventory	5
1.1	Accessible sources and how they were used	5
1.2	Pattern analysis from the sources	7
2	Detailed blueprint	7
2.1	Module objective	7
2.2	Prerequisite concepts	8
2.3	Canonical models	8
2.4	Exam variations	8
2.5	Closed-form catalog to master	9
2.6	Exercise plan	9
3	Module map	9
4	Learning objectives	10
5	Static optimization toolkit	11
5.1	Unconstrained problems	11
5.2	Equality constraints	11
5.3	Inequality constraints and KKT conditions	12
5.4	Envelope theorem	12
5.5	Comparative statics through implicit differentiation	13
6	Recursive formulation workflow	13
6.1	The five questions that build every Bellman equation	13
6.2	Sequential to recursive: standard derivation	14
7	Dynamic programming theory	14
7.1	Bellman operator	14
7.2	Blackwell sufficient conditions	14
7.3	Bounded state restrictions	15
7.4	Monotonicity, concavity, and single-valued policies	15
7.5	FOC-envelope-Euler workflow	15
8	Canonical model 1: static quadratic reaching the center	16
8.1	Static problem	16
8.2	Finite horizon induction	16
8.3	Infinite horizon	17
9	Canonical model 2: tree-cutting and stopping	17
9.1	Environment and Bellman equation	17
9.2	Contraction on the useful function space	18
9.3	Cutoff logic	18

10 Canonical model 3: deterministic optimal growth	18
10.1 Environment	18
10.2 Recursive formulation	19
10.3 FOC, envelope, and Euler	19
10.4 KKT version	19
11 Canonical model 4: growth with leisure	20
11.1 Environment	20
11.2 Mapping into $v(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}$	20
11.3 Labor-leisure FOC	20
11.4 Conditions for F to be well defined and increasing	20
11.5 Strict concavity of the value function	21
12 Canonical model 5: stochastic optimal growth	21
12.1 Environment	21
12.2 Bellman equation	21
12.3 Bounded state restriction	21
12.4 Existence and uniqueness	21
12.5 Concavity, monotonicity, and policy	22
12.6 FOC and Euler equation	22
12.7 Why $g(k, z) = h(zk^\alpha)$	22
13 Closed-form derivation lab	23
13.1 General guess-and-verify checklist	23
13.2 Log cake-eating	23
13.3 Log Cobb-Douglas growth	23
13.4 Log AK growth	24
13.5 Natural resource extraction	24
13.6 Human capital accumulation	25
13.7 Survival-risk project choice	26
13.8 Disaster-risk AK with CRRA utility	27
13.9 Alternating management rights	27
14 Closed-form catalog	28
15 Core derivation templates	29
15.1 Template A: static KKT	29
15.2 Template B: recursive formulation	29
15.3 Template C: FOC-envelope-Euler	30
15.4 Template D: coefficient matching	30
16 Exam variations	30
16.1 What changes and what does not	30
16.2 Past-qual practice map	31
17 Common traps	31

18 Exercise bank	32
18.1 Mechanical drills	32
18.2 Closed-form and proof drills	32
18.3 Past-qual-style applications	33
18.4 Trap and Economist X questions	33
19 Answer sketches	34
19.1 Mechanical drills	34
19.2 Closed-form and proof drills	34
19.3 Past-qual-style applications	35
19.4 Trap and Economist X questions	35
20 Final study checklist	35
Quality audit	36

1 Source inventory

1.1 Accessible sources and how they were used

The packet follows the project hierarchy: past ASU qualifying exams determine exam relevance, PS1 determines current-course emphasis, and the manifest/skill determine structure and quality gates. No separate Galina lecture-note or slide files were accessible in the provided working directory beyond the uploaded PS and qual files. I therefore treat the uploaded 2016–2024 qualifying-exam .tex files as the past-qual backbone and PS1 as the current dynamic-programming anchor.

Source	Type	Module 1 topics	Math tools	Use / issue
MACRO_MANIFEST.md	Project manifest	Locked Module 1 requirements; source hierarchy; PS1 anchor; required derivations; output standards	Packet architecture and topic coverage	Primary steering document; no extraction issue.
SKILL.md	Builder skill	Required workflow; source inventory; pattern analysis; blueprint; canonical model structure; exercise standards; TeX rules	Packet design and audit checklist	Primary methodology document; no extraction issue.
PS1-5.pdf	Current 2026 problem set	Reaching-the-center quadratic problem; tree-cutting; growth with leisure; stochastic growth; bounded state space; contraction; concavity; monotonicity; single-valued policies	Static optimization; finite and infinite horizon DP; Blackwell/contraction; bounded DP; stochastic Bellman	Core current-course anchor. PDF text extraction was readable; notation cleaned in the packet.
2016-06_...tex	Past qual	Multi-household planner growth; aggregate capital dependence; Ramsey/growth with taxes	Planner FOCs; Pareto weights; aggregate state reduction; implementability logic	High relevance for planner setup and aggregate-state reasoning; Mathpix-like numbering irregularities.

Source	Type	Module 1 topics	Math tools	Use / issue
2017-06_...tex	Past qual	Endogenous growth with externality/-government policy; no-capital optimal labor taxation	Sequential equilibrium; planner FOCs; policy wedges; static FOCs	Moderate Module 1 relevance; mostly equilibrium/welfare cross-over.
2018-06_...tex	Past qual	Natural resource extraction with growing stock; log closed form; steady-state condition	Sequential setup; Euler equation; log guess-and-verify	High relevance for creative closed-form dynamic problems.
2019-06_...tex	Past qual	Irreversible work/non-work choice with saving; borrowing constraint; policy-function comparison	Dynamic decision setup; Euler equation; cutoff in initial assets; constraint logic	Moderate-high relevance for state choice and KKT/non-negativity constraints.
2020-06_...tex	Past qual	Growth with endogenous labor and public leisure; recursive planner problem; FOC/envelope; steady state; log Cobb-Douglas hours	Recursive formulation; labor-leisure FOCs; envelope; steady-state closed form	High relevance for growth-with-leisure and public externality warnings.
2021-06_...tex	Past qual	Skill accumulation; recursive setup; Euler equation; log guess-and-verify linear investment	Bellman; Euler; co-efficient matching; comparative statics	High relevance for closed-form policy derivation.
2022-06_...tex	Past qual	Safe/risky project with survival probability; insurance after failure; cutoff project choice	Recursive setup; log value guess; cutoff proof; policy comparison	High relevance for dynamic project choice and cutoff derivations.

Source	Type	Module 1 topics	Math tools	Use / issue
2023-06_...tex	Past qual	AK production; independent management versus joint asset with alternating management rights	Recursive setup; log AK saving; linear-policy equilibrium; strategic recursion	High relevance for creative closed-form recursive problems.
2024-06_...tex	Past qual	Disaster-risk AK model; recursive planner; Euler; CRRA optimal saving rule; comparative statics	Stochastic Euler; CRRA closed form; precautionary/re-turn effects	High relevance and most recent Galina-style dynamic problem.

1.2 Pattern analysis from the sources

The repeated exam command is not merely “solve the model.” The command is usually one of the following:

1. **Formulate recursively:** name the state, control, feasible set, timing, and Bellman equation.
2. **Derive FOC and envelope:** state interiority or KKT conditions, then use the envelope theorem cleanly.
3. **Derive an Euler equation:** eliminate the derivative of the value function using the next-period envelope condition.
4. **Guess and verify:** choose the right value-function shape, substitute, match coefficients, recover the policy, and verify feasibility.
5. **Prove a property:** contraction, boundedness, monotonicity, concavity, single-valuedness, cut-off/stopping form.
6. **Answer an Economist X claim:** use the formal equations first, then give intuition; never rely on intuition alone.

Exam-facing rule

For Module 1, the exam reward is usually not from remembering a theorem statement. It is from writing the right state and transition and then moving fluently through

Bellman $\implies$ FOC/KKT $\implies$ Envelope $\implies$ Euler or policy rule.

2 Detailed blueprint

2.1 Module objective

By the end of this module, a student should be able to take an unfamiliar dynamic macro problem and produce a grader-readable solution in real time. The student should be able to decide whether

the problem is a static optimization, deterministic DP, stochastic DP, finite-horizon induction, stopping problem, growth model, or closed-form recursive problem; then derive the relevant conditions without losing timing or state variables.

2.2 Prerequisite concepts

- Multivariable calculus: gradients, Hessians, implicit differentiation.
- Concavity and convexity: why concavity turns FOCs into sufficient conditions for maximization.
- Basic probability: conditional expectation and Markov transition kernels.
- Macro feasibility: output can be consumed or invested; capital laws of motion; labor-leisure time constraints.
- Comfort with logs and CRRA algebra.

2.3 Canonical models

1. Static optimization and KKT conditions.
2. Abstract deterministic and stochastic Bellman equations.
3. Finite-horizon induction and the quadratic reaching-the-center problem.
4. Infinite-horizon quadratic guess-and-verify.
5. Tree-cutting / stopping.
6. Deterministic optimal growth: Bellman, FOC, envelope, Euler.
7. Growth with leisure: mapping to $v(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}$.
8. Stochastic optimal growth: bounded state, contraction, monotonicity, concavity, single-valued policy.
9. Log cake-eating, log Cobb-Douglas growth, and log AK growth.
10. Creative closed forms: natural resource extraction, human capital accumulation, survival-risk project choice, disaster-risk AK, and alternating management rights.

2.4 Exam variations

The same skeleton appears in different clothing:

- The state can be a physical stock, a productivity shock, an employment status, a project survival status, a human-capital level, or a vector of sector-specific skills.
- A control can be consumption, saving, next capital, labor, leisure, investment in skill, project choice, or stopping.
- A stochastic Bellman requires a conditional expectation over next states; a deterministic Bellman does not.
- A closed-form problem usually has homogeneity. Look for log utility, Cobb-Douglas or linear technology, and multiplicative transitions.

- A cutoff problem usually compares two value functions whose difference is monotone in the state.

2.5 Closed-form catalog to master

Problem	Correct guess	High-yield policy result
Quadratic reaching-the-center	$v(x) = Ax^2$	$x' = x/(2 + \beta A)$ and $A = (1 + \beta A)/(2 + \beta A)$.
Log cake-eating	$v(x) = a + b \log x$	$c = (1 - \beta)x$, $x' = \beta x$.
Log Cobb-Douglas growth	$v(k) = a + b \log k$	$k' = \alpha \beta A k^\alpha$.
Log AK growth	$v(k) = a + b \log k$	$k' = \beta A k$, $c = (1 - \beta) A k$.
Natural resource, $u = \log c$, $f(l) = l^\gamma$	$v(x) = a + b \log x$	$l = (1 - \beta)x$, $x' = \beta(1 + g)x$.
Human capital with log utility	$v(s) = a + b \log s$, $i = js$	j solves a scalar FOC; investment is linear in skill.
Survival project with log utility	$V_i(k) = a_i + b_i \log k$	Conditional on project i , $k' = \beta \gamma_i k$ and project choice has a cutoff in k_0 .
Disaster-risk AK with CRRA	Homothetic value $V(k) = Dk^{1-\sigma} + C$	Saving share $s = [\beta A^{1-\sigma}((1 - q) + q\rho^{1-\sigma})]^{1/\sigma}$.
Alternating management rights	$V(k) = a + b \log k$, $g(k) = \eta A k$	Joint asset evolves as $K' = \beta^2 A K$, slower than independent management $K' = \beta A K$.

2.6 Exercise plan

The exercise bank at the end contains four categories: mechanical drills, closed-form/proof drills, past-qual-style applications, and trap/Economist X claims. The intended order is: first complete all mechanical drills without looking at sketches; then do closed-form drills with coefficient matching; then attempt past-qual applications under a 20-minute clock.

3 Module map

Subtopic	Why it matters	Source appearances	appear-	Core technique	Common trap
KKT and Lagrangians	Converts constraints into conditions and multipliers	Static part of PS1; household/planner FOCs across quals		Lagrangian, complementary slackness, concavity sufficiency	Ignoring inequality constraints.

Subtopic	Why it matters	Source ances	appear-	Core technique	Common trap
Envelope theorem	Needed for Euler equations and value derivatives	PS1; 2020 Q1; 2021 Q3; 2024 Q1		Differentiate maximized value w.r.t. state	Differentiating the policy.
Recursive formulation	First step in most modern questions	PS1; 2018 Q3; 2020 Q1; 2021 Q3; 2022 Q1; 2023 Q3; 2024 Q1		State-control-transition-feasible set	Putting controls in the state.
Contraction mapping	Proves unique Bellman solution	PS1 tree-cutting and stochastic growth		Blackwell monotonicity and discounting	Forgetting bounded state space.
Bounded stochastic growth	Current PS1 emphasis	PS1 Q4		Choose invariant compact K ; apply contraction	Letting k be unbounded.
Concavity and monotone policies	Proves value/policy properties	PS1 Q3/Q4		Concavity preservation, strict concavity, monotone comparative statics	Claiming strict concavity without strict primitives.
FOC-envelope-Euler	Highest-value dynamic derivation	Growth, resources, human capital, disaster AK		Combine current FOC and next envelope	Timing the marginal product wrong.
Guess-and-verify	Closed-form exam engine	PS1 Q1; 2018 Q3; 2021 Q3; 2022 Q1; 2023 Q3; 2024 Q1		Guess, substitute, match coefficients, verify	Guessing policy before value.
Stopping/cutoff	Tree and project choices	PS1 tree-cutting; 2022 Q1		Compare continuation vs stopping/project values	Not proving monotonicity.
Economist claims	Common final subpart style	2019–2024 quals		Formal sign/-comparison plus intuition	Pure intuition with no equation.

4 Learning objectives

After studying this packet, you should be able to do the following under exam conditions:

1. Given a static maximization or minimization problem, write the Lagrangian or KKT system and state when the conditions are sufficient.

2. Given a sequential dynamic problem, identify the minimal state vector and write the Bellman equation with the correct feasible correspondence.
3. Given a differentiable Bellman problem, derive the FOC, envelope condition, and Euler equation without timing mistakes.
4. Given a bounded stochastic growth problem, construct a compact invariant state space and prove the Bellman operator has a unique bounded continuous fixed point.
5. Given a concave dynamic problem, prove that the value function is increasing/concave and the policy is single-valued and continuous.
6. Given a log or CRRA dynamic problem, choose an appropriate value-function guess and derive a closed-form policy by coefficient matching.
7. Given a stopping or project-choice problem, prove a cutoff or threshold rule by comparing values and establishing monotonicity.
8. Given an Economist X claim, translate the claim into a sign statement and resolve it formally before giving intuition.

5 Static optimization toolkit

5.1 Unconstrained problems

For

$$\max_{x \in X \subseteq \mathbb{R}^n} f(x),$$

if x^* is interior and f is differentiable, the necessary condition is

$$\nabla f(x^*) = 0.$$

If f is concave on a convex feasible set, any point satisfying the first-order condition is a global maximum. If f is strictly concave, the maximizer is unique. For minimization, replace concavity with convexity.

Qualifier trap

FOCs are not automatically sufficient. In a maximization problem, you need concavity of the objective and convexity of the feasible set, or you need to verify second-order/global conditions directly.

5.2 Equality constraints

Consider

$$\max_x f(x) \quad \text{s.t.} \quad g_j(x) = 0, \quad j = 1, \dots, m.$$

Use the Lagrangian

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{j=1}^m \lambda_j g_j(x).$$

The first-order conditions are

$$\nabla_x f(x^*) + \sum_{j=1}^m \lambda_j \nabla_x g_j(x^*) = 0, \quad g_j(x^*) = 0.$$

The multiplier λ_j is the marginal value of relaxing the constraint, after sign conventions are made precise. If the constraint is written as resource available minus resource used, a positive multiplier usually means an extra unit of the resource raises the objective.

5.3 Inequality constraints and KKT conditions

A reusable template for a maximization problem is

$$\max_x f(x) \quad \text{s.t.} \quad h_i(x) \geq 0, \quad i = 1, \dots, m.$$

Write

$$\mathcal{L}(x, \mu) = f(x) + \sum_{i=1}^m \mu_i h_i(x), \quad \mu_i \geq 0.$$

The KKT conditions are

$$\begin{aligned} \nabla_x f(x^*) + \sum_{i=1}^m \mu_i \nabla_x h_i(x^*) &= 0, & (\text{stationarity}) \\ h_i(x^*) &\geq 0, & (\text{primal feasibility}) \\ \mu_i &\geq 0, & (\text{dual feasibility}) \\ \mu_i h_i(x^*) &= 0. & (\text{complementary slackness}) \end{aligned}$$

If f is concave and every h_i is concave, the feasible set $\{x : h_i(x) \geq 0\}$ is convex and KKT conditions are sufficient under standard constraint qualifications.

Reusable formula

Economic interpretation. If a nonnegativity constraint $k' \geq 0$ has multiplier $\mu > 0$, the planner wants to choose less than zero but is prevented. If a resource constraint $f(k) - c - k' \geq 0$ has multiplier $\lambda > 0$, one more unit of resources increases value by λ .

5.4 Envelope theorem

Let

$$V(x) = \max_{y \in \Gamma(x)} F(x, y).$$

If $y^*(x)$ is an interior differentiable policy and regularity conditions hold, then

$$V'(x) = F_x(x, y^*(x)).$$

You do not include $F_y y'(x)$ because $F_y = 0$ at an interior optimum. With constraints, include the direct effect of the state on the Lagrangian, evaluated at the optimum:

$$V_x(x) = \mathcal{L}_x(x, y^*(x), \lambda^*(x)).$$

Qualifier trap

In dynamic programming, the envelope condition differentiates with respect to the *state*, not the control. If the Bellman is $v(k) = \max_{k'} u(f(k) - k') + \beta v(k')$, then $v'(k) = u'(c)f'(k)$, not $u'(c)f'(k) + \beta v'(k')g'(k)$.

5.5 Comparative statics through implicit differentiation

Suppose an interior choice $x^*(\theta)$ solves

$$F(x, \theta) = 0.$$

If $F_x \neq 0$, then

$$\frac{dx^*}{d\theta} = -\frac{F_\theta(x^*, \theta)}{F_x(x^*, \theta)}.$$

On the exam, sign the denominator using the second-order condition. For a maximization problem with scalar choice, the FOC is $F = f_x = 0$ and the SOC says $F_x = f_{xx} < 0$. Hence the sign of $dx^*/d\theta$ is the sign of F_θ .

6 Recursive formulation workflow**6.1 The five questions that build every Bellman equation****Workflow**

1. **What is predetermined at the decision date?** These are state variables.
2. **What does the agent choose now?** These are controls.
3. **What are the feasibility constraints?** This gives $\Gamma(x)$.
4. **How does today plus choices determine tomorrow?** This is the transition.
5. **What is the current payoff and what is discounted future value?** This gives the Bellman equation.

A deterministic stationary Bellman equation usually has the form

$$v(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}.$$

A stochastic stationary Bellman equation usually has the form

$$v(x, z) = \max_{y \in \Gamma(x, z)} \{F(x, z, y) + \beta \mathbb{E} [v(y, z') \mid z]\}.$$

If the shock is realized before the choice, it belongs in the current state. If it is realized after the choice, it belongs in the transition expectation.

6.2 Sequential to recursive: standard derivation

Start from a sequential problem:

$$\begin{aligned} \max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & c_t + k_{t+1} \leq f(k_t) + (1 - \delta)k_t, \\ & c_t \geq 0, \quad k_{t+1} \geq 0, \quad k_0 \text{ given.} \end{aligned}$$

The payoff is additively separable over time, k_t summarizes all payoff-relevant history, and the transition from k_t to k_{t+1} is Markov. Therefore define

$$\Gamma(k) = \{k' \geq 0 : k' \leq f(k) + (1 - \delta)k\}.$$

Substituting $c = f(k) + (1 - \delta)k - k'$ gives

$$v(k) = \max_{k' \in \Gamma(k)} \{u(f(k) + (1 - \delta)k - k') + \beta v(k')\}.$$

Exam-facing rule

A state variable is not “anything that appears in the problem.” It is a variable that is predetermined at the decision date and needed to evaluate current payoffs or future feasible sets. A control is chosen at the decision date. A shock observed before choice is a state; a shock drawn after choice enters the expectation.

7 Dynamic programming theory

7.1 Bellman operator

For a bounded continuous function v , define the Bellman operator

$$(Tv)(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}.$$

A solution to the Bellman equation is a fixed point $v = Tv$. The policy correspondence is

$$G(x) = \arg \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}.$$

7.2 Blackwell sufficient conditions

Let $\mathcal{B}(X)$ be the space of bounded functions with sup norm. Suppose T maps $\mathcal{B}(X)$ into itself. Blackwell's sufficient conditions are:

1. **Monotonicity:** if $v \leq w$, then $Tv \leq Tw$.
2. **Discounting:** for any constant $a \geq 0$,

$$T(v + a) \leq Tv + \beta a.$$

Then T is a contraction with modulus β :

$$\|Tv - Tw\|_{\infty} \leq \beta \|v - w\|_{\infty}.$$

By the contraction mapping theorem, T has a unique fixed point and value iteration converges to it from any starting bounded function.

7.3 Bounded state restrictions

Unbounded state spaces often prevent direct use of bounded DP tools. In optimal growth, restrict to a compact invariant set. For the PS1 stochastic growth problem with full depreciation and production zk^α , $z \in [\underline{z}, \bar{z}]$, choose

$$\bar{k} \geq \max \left\{ k_0, 1, \bar{z}^{1/(1-\alpha)} \right\}.$$

Then for any $k \in [0, \bar{k}]$ and $z \leq \bar{z}$,

$$0 \leq k' \leq zk^\alpha \leq \bar{z}\bar{k}^\alpha \leq \bar{k}.$$

Thus $K = [0, \bar{k}]$ is invariant.

Qualifier trap

Do not write “ T is a contraction” before checking the space. In stochastic growth, $u(c)$ may be unbounded near $c = 0$ and capital may be unbounded unless you impose an invariant compact state space and handle boundary issues carefully.

7.4 Monotonicity, concavity, and single-valued policies

A standard proof template:

1. If $F(x, y)$ is increasing in x and Γ is monotone in x , then T maps increasing functions into increasing functions. Starting value iteration from an increasing function gives an increasing fixed point.
2. If F is concave and the graph of Γ is convex, then T maps concave functions into concave functions. The fixed point is concave.
3. If the objective $F(x, y) + \beta v(y)$ is strictly concave in y on a convex feasible set, the policy correspondence is single-valued.
4. If the objective is continuous and Γ is nonempty, compact-valued, and continuous, the theorem of the maximum gives continuity of v and upper hemicontinuity of G . If G is single-valued, the policy function is continuous.

7.5 FOC-envelope-Euler workflow

For

$$v(k) = \max_{k'} \{u(f(k) + (1 - \delta)k - k') + \beta v(k')\},$$

let $c = f(k) + (1 - \delta)k - k'$. The interior FOC is

$$u'(c) = \beta v'(k').$$

The envelope condition is

$$v'(k) = u'(c) [f'(k) + 1 - \delta].$$

Evaluate the envelope condition next period:

$$v'(k_{t+1}) = u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta].$$

Substitute into the FOC:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta].$$

Reusable formula

Euler equation in words. The marginal utility cost of saving one more unit today equals the discounted marginal utility benefit tomorrow times the gross marginal product of capital tomorrow.

8 Canonical model 1: static quadratic reaching the center

8.1 Static problem

An agent begins at $x_0 \in [-1, 1]$ and chooses $x \in [-1, 1]$ to minimize travel cost plus distance cost:

$$\min_{x \in [-1, 1]} (x - x_0)^2 + x^2.$$

The FOC is

$$2(x - x_0) + 2x = 0,$$

so

$$x^* = \frac{x_0}{2}.$$

Since $x_0 \in [-1, 1]$, $x_0/2 \in [-1, 1]$, so the constraint does not bind. The objective is strictly convex, so the FOC is sufficient.

8.2 Finite horizon induction

Let $v_n(x)$ be the minimized cost with n periods remaining. With one period left,

$$v_1(x) = \min_y \{(y - x)^2 + y^2\} = \frac{1}{2}x^2.$$

Guess $v_{n-1}(x) = A_{n-1}x^2$. Then

$$\begin{aligned} v_n(x) &= \min_y \{(y - x)^2 + y^2 + \beta A_{n-1}y^2\} \\ &= \min_y \{(y - x)^2 + (1 + \beta A_{n-1})y^2\}. \end{aligned}$$

The FOC is

$$2(y - x) + 2(1 + \beta A_{n-1})y = 0,$$

so

$$y_n(x) = \frac{x}{2 + \beta A_{n-1}}.$$

The minimized value is

$$A_n x^2 = \frac{1 + \beta A_{n-1}}{2 + \beta A_{n-1}} x^2,$$

hence

$$A_n = \frac{1 + \beta A_{n-1}}{2 + \beta A_{n-1}}, \quad A_1 = \frac{1}{2}.$$

This proves by induction that $v_n(x) = A_n x^2$.

8.3 Infinite horizon

The Bellman equation is

$$v(x) = \min_y \{ (y - x)^2 + y^2 + \beta v(y) \}.$$

Guess $v(x) = Ax^2$ with $A \geq 0$. Then

$$v(x) = \min_y \{ (y - x)^2 + (1 + \beta A)y^2 \}.$$

The policy is

$$y(x) = \frac{x}{2 + \beta A}.$$

The coefficient must satisfy

$$A = \frac{1 + \beta A}{2 + \beta A}.$$

Equivalently,

$$\beta A^2 + (2 - \beta)A - 1 = 0.$$

The nonnegative root is

$$A = \frac{\beta - 2 + \sqrt{\beta^2 + 4}}{2\beta}, \quad \beta > 0,$$

with limiting value $A = 1/2$ when $\beta = 0$. The other root is negative. Therefore the nonnegative solution is unique. The state evolves according to

$$x_{t+1} = mx_t, \quad m = \frac{1}{2 + \beta A} \in (0, 1),$$

so

$$x_t = m^t x_0 \rightarrow 0.$$

Exam-facing rule

This is the cleanest template for quadratic guess-and-verify: guess quadratic, substitute, solve the linear policy from the FOC, plug back, and solve a scalar coefficient equation.

9 Canonical model 2: tree-cutting and stopping

9.1 Environment and Bellman equation

A tree of size x grows according to $x' = h(x)$ if not cut. If cut today, payoff is x . The owner chooses a cutting time to maximize $\beta^T x_T$. Normalize the price of wood to one.

The state is current tree size x . The control is binary: cut or wait. The Bellman equation is

$$v(x) = \max\{x, \beta v(h(x))\}.$$

The stopping region is

$$\mathcal{S} = \{x : x \geq \beta v(h(x))\}.$$

The continuation region is

$$\mathcal{C} = \{x : x < \beta v(h(x))\}.$$

9.2 Contraction on the useful function space

Let A be the set of continuous nondecreasing functions satisfying $v(x) = x$ for all $x \geq \bar{x}$, where $\beta h(x)/x < 1$ for $x > \bar{x}$. For $v \in A$,

$$(Tv)(x) = \max\{x, \beta v(h(x))\}.$$

If $x \geq \bar{x}$ and $h(x) \geq \bar{x}$, then $v(h(x)) = h(x)$, so $\beta v(h(x)) = \beta h(x) < x$. If $x \geq \bar{x}$ and $h(x) < \bar{x}$, monotonicity gives $v(h(x)) \leq v(\bar{x}) = \bar{x} \leq x$, hence $\beta v(h(x)) \leq x$. Thus $(Tv)(x) = x$ for $x \geq \bar{x}$. Continuity and monotonicity are inherited from the maximum of continuous nondecreasing functions. Therefore T maps A into A .

Use the norm

$$\|v - w\| = \sup_{x \in [0, \bar{x}]} |v(x) - w(x)|.$$

For $x \geq \bar{x}$, all functions in A coincide, so the difference is zero. For $x \in [0, \bar{x}]$,

$$|Tv(x) - Tw(x)| \leq \beta \|v - w\|.$$

Thus T is a contraction on the closed complete space $(A, \|\cdot\|)$ and has a unique fixed point.

9.3 Cutoff logic

Suppose there is x^* such that

$$\frac{\beta h(x)}{x} > 1 \quad \text{if } x < x^*, \quad \frac{\beta h(x)}{x} < 1 \quad \text{if } x > x^*.$$

The one-period discounted value of waiting exceeds cutting below x^* and falls short above x^* . Under the maintained tree-growth monotonicity assumptions, the optimal rule is to wait while $x < x^*$ and cut once $x \geq x^*$.

The exam proof should say:

1. If $x > x^*$, then one-period waiting gives less discounted size than cutting. Repeated waiting while above the threshold only compounds this loss, so cutting is optimal.
2. If $x < x^*$, then waiting at least one period raises discounted size relative to cutting. Continue this argument until the first time the state reaches the stopping region.

Qualifier trap

Do not write a cutoff rule without identifying the object being compared. In tree-cutting, compare x and $\beta v(h(x))$. In project choice, compare $V_S(k)$ and $V_R(k)$. In employment choice, compare $W(a)$ and $N(a)$.

10 Canonical model 3: deterministic optimal growth

10.1 Environment

A representative planner maximizes

$$\sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} \leq f(k_t) + (1 - \delta)k_t, \quad c_t \geq 0, \quad k_{t+1} \geq 0,$$

with k_0 given. Assume u is increasing, strictly concave, differentiable, and satisfies an Inada condition; f is increasing, concave, differentiable.

10.2 Recursive formulation

State: k . Control: k' or equivalently c . Feasible correspondence:

$$\Gamma(k) = \{k' \geq 0 : k' \leq f(k) + (1 - \delta)k\}.$$

Bellman equation:

$$v(k) = \max_{k' \in \Gamma(k)} \{u(f(k) + (1 - \delta)k - k') + \beta v(k')\}.$$

10.3 FOC, envelope, and Euler

Let

$$c = f(k) + (1 - \delta)k - k'.$$

Assuming an interior solution, the FOC is

$$u'(c) = \beta v'(k').$$

The envelope condition is

$$v'(k) = u'(c) [f'(k) + 1 - \delta].$$

Therefore the Euler equation is

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta].$$

10.4 KKT version

If $k' \geq 0$ can bind, write

$$\mathcal{L} = u(c) + \beta v(k') + \lambda [f(k) + (1 - \delta)k - c - k'] + \mu k'.$$

FOCs:

$$\begin{aligned} u'(c) - \lambda &= 0, \\ \beta v'(k') - \lambda + \mu &= 0, \\ \mu &\geq 0, \quad k' \geq 0, \quad \mu k' = 0. \end{aligned}$$

Thus

$$u'(c) \geq \beta v'(k'),$$

with equality when $k' > 0$.

Exam-facing rule

If the problem says Inada and $k_0 > 0$, you can often state that the consumption choice is interior. But if a borrowing/nonnegativity constraint is central to the question, use KKT

and do not hide the inequality.

11 Canonical model 4: growth with leisure

11.1 Environment

The planner values consumption and leisure. Let n denote labor and $\ell = 1 - n$ denote leisure. The per-period utility is $U(c, 1 - n)$ and production is $f(k, n)$. Feasibility is

$$c + k' \leq f(k, n) + (1 - \delta)k, \quad 0 \leq n \leq 1, \quad k' \geq 0.$$

11.2 Mapping into $v(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}$

Use $x = k$ and $y = k'$. The feasible correspondence for next capital is

$$\Gamma(k) = \{k' \geq 0 : \exists n \in [0, 1] \text{ with } f(k, n) + (1 - \delta)k - k' \geq 0\}.$$

Define the one-period return after optimizing over labor:

$$F(k, k') = \max_{n \in [0, 1]} U(f(k, n) + (1 - \delta)k - k', 1 - n),$$

where the domain requires nonnegative consumption. Then

$$v(k) = \max_{k' \in \Gamma(k)} \{F(k, k') + \beta v(k')\}.$$

11.3 Labor-leisure FOC

If $n \in (0, 1)$ and $c > 0$, the intratemporal FOC is

$$U_c(c, 1 - n)f_n(k, n) = U_\ell(c, 1 - n).$$

The left side is the utility value of the extra output from one more unit of labor. The right side is the marginal utility cost of losing leisure.

11.4 Conditions for F to be well defined and increasing

Sufficient conditions:

- U is continuous on the relevant positive consumption/leisure domain and strictly increasing in both arguments.
- f is continuous and increasing in k .
- The choice set for n is compact, $[0, 1]$.
- For each feasible (k, k') , there exists some n with positive consumption.

Then the theorem of the maximum gives continuity of F . Since a higher k raises feasible consumption for any fixed n and U is increasing in consumption, F is increasing in k .

11.5 Strict concavity of the value function

Sufficient conditions:

- U is strictly concave in (c, ℓ) .
- f is concave in (k, n) and strictly concave in at least one relevant dimension or combined with strict concavity of U to make the induced return strictly concave.
- The feasible graph is convex.
- $\beta \in (0, 1)$.

Under these conditions, T maps concave functions into concave functions. Strict concavity follows if mixing two distinct feasible plans strictly improves the current return. The fixed point is strictly concave, and the policy is single-valued when the maximand is strictly concave in controls.

12 Canonical model 5: stochastic optimal growth

12.1 Environment

Utility is $U(c)$, production is $zk^\alpha n^{1-\alpha}$, the agent has one unit of time and does not value leisure, and capital fully depreciates. Thus $n = 1$ and output is zk^α . The shock z is iid with support $Z = [\underline{z}, \bar{z}]$ and distribution λ .

12.2 Bellman equation

The state is (k, z) . The control is k' . Consumption is

$$c = zk^\alpha - k'.$$

The feasible correspondence is

$$\Gamma(k, z) = \{k' \in [0, \bar{k}] : 0 \leq k' \leq zk^\alpha\}.$$

The Bellman equation is

$$v(k, z) = \max_{0 \leq k' \leq zk^\alpha} \left\{ U(zk^\alpha - k') + \beta \int_Z v(k', z') \lambda(dz') \right\}.$$

12.3 Bounded state restriction

Choose $\bar{k} \geq \max\{k_0, 1, \bar{z}^{1/(1-\alpha)}\}$. Then $zk^\alpha \leq \bar{k}$ whenever $k \in [0, \bar{k}]$ and $z \in Z$. Hence $K = [0, \bar{k}]$ is invariant.

12.4 Existence and uniqueness

Let $\mathcal{C}(K \times Z)$ be the space of bounded continuous functions. If U is continuous and bounded on the feasible consumption domain after handling the boundary in the usual way, and if the feasible correspondence is nonempty, compact-valued, and continuous, then the Bellman operator maps $\mathcal{C}(K \times Z)$ into itself. Blackwell's conditions imply it is a contraction:

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty.$$

Therefore a unique bounded continuous value function exists.

12.5 Concavity, monotonicity, and policy

For fixed z , the current return $U(zk^\alpha - k')$ is strictly concave in (k, k') on the interior because U is strictly concave and k^α is concave. The continuation value is concave in k' if v is concave. Therefore T preserves concavity and the fixed point is strictly concave in k . Since the maximand is strictly concave in k' , the policy correspondence is single-valued. Continuity follows from the theorem of the maximum.

12.6 FOC and Euler equation

Assuming an interior solution,

$$U'(c) = \beta \int_Z v_k(k', z') \lambda(dz').$$

The envelope condition is

$$v_k(k, z) = U'(c) \alpha z k^{\alpha-1}.$$

Thus

$$U'(c_t) = \beta \int_Z U'(c_{t+1}(z')) \alpha z' (k_{t+1})^{\alpha-1} \lambda(dz').$$

12.7 Why $g(k, z) = h(zk^\alpha)$

Define current resources

$$y = zk^\alpha.$$

The Bellman problem can be written

$$v(k, z) = \max_{0 \leq k' \leq y} \left\{ U(y - k') + \beta \int_Z v(k', z') \lambda(dz') \right\}.$$

The right-hand side depends on (k, z) only through y . Therefore the optimal next capital can be written as

$$g(k, z) = h(y) = h(zk^\alpha).$$

Strict concavity and increasing differences imply h is continuous and strictly increasing in y . Since $y = zk^\alpha$ is strictly increasing in both k and z , g is strictly increasing in both arguments. The consumption policy is

$$c(k, z) = zk^\alpha - h(zk^\alpha).$$

Using the FOC, higher y raises both optimal saving and optimal consumption; otherwise marginal utility and marginal continuation value would move in contradictory directions. Hence $c(k, z)$ is also increasing in both arguments.

Qualifier trap

The current resource variable $y = zk^\alpha$ is not the state of the original stochastic process, but it is the sufficient scalar for the current choice because shocks are iid and continuation value depends on k' only through next-period capital.

13 Closed-form derivation lab

13.1 General guess-and-verify checklist

Workflow

1. Inspect homogeneity. Logs usually suggest $v(x) = a + b \log x$. Quadratics suggest $v(x) = Ax^2$. CRRA with AK suggests $v(k) = Dk^{1-\sigma} + C$.
2. Write the Bellman equation with the state and control.
3. Substitute the guess before solving the policy.
4. Derive the FOC and solve the policy as a function of the unknown coefficient.
5. Plug the policy back into the Bellman equation.
6. Match coefficients on $\log x$, x^2 , or $x^{1-\sigma}$.
7. Check feasibility, interiority, transversality, and uniqueness.

13.2 Log cake-eating

The problem is

$$v(x) = \max_{0 \leq c \leq x} \{\log c + \beta v(x - c)\}.$$

Guess $v(x) = a + b \log x$. Then

$$a + b \log x = \max_c \{\log c + \beta a + \beta b \log(x - c)\}.$$

FOC:

$$\frac{1}{c} = \frac{\beta b}{x - c}.$$

Thus

$$c = \frac{x}{1 + \beta b}, \quad x' = x - c = \frac{\beta b}{1 + \beta b} x.$$

Match the coefficient on $\log x$:

$$b = 1 + \beta b \implies b = \frac{1}{1 - \beta}.$$

Therefore

$$c = (1 - \beta)x, \quad x' = \beta x.$$

13.3 Log Cobb-Douglas growth

Let

$$v(k) = \max_{0 \leq k' \leq Ak^\alpha} \{\log(Ak^\alpha - k') + \beta v(k')\}.$$

Guess $v(k) = a + b \log k$. FOC:

$$\frac{1}{Ak^\alpha - k'} = \frac{\beta b}{k'}.$$

Hence

$$k' = \frac{\beta b}{1 + \beta b} Ak^\alpha.$$

Match coefficients:

$$b = \alpha(1 + \beta b) \implies b = \frac{\alpha}{1 - \alpha\beta}.$$

Then

$$\frac{\beta b}{1 + \beta b} = \alpha\beta,$$

so

$$k' = \alpha\beta Ak^\alpha, \quad c = (1 - \alpha\beta)Ak^\alpha.$$

13.4 Log AK growth

The Bellman equation is

$$v(k) = \max_{0 \leq k' \leq Ak} \{\log(Ak - k') + \beta v(k')\}.$$

Guess $v(k) = a + b \log k$. The coefficient condition is

$$b = 1 + \beta b,$$

so $b = 1/(1 - \beta)$. The FOC gives

$$k' = \frac{\beta b}{1 + \beta b} Ak = \beta Ak,$$

so

$$c = (1 - \beta)Ak.$$

13.5 Natural resource extraction

The sequential problem is

$$\max_{\{l_t\}} \sum_{t=0}^{\infty} \beta^t u(f(l_t))$$

subject to

$$0 \leq l_t \leq x_t, \quad x_{t+1} = (x_t - l_t)(1 + g).$$

The Bellman equation is

$$v(x) = \max_{0 \leq l \leq x} \{u(f(l)) + \beta v((x - l)(1 + g))\}.$$

The Euler equation for an interior allocation is

$$u'(f(l_t))f'(l_t) = \beta(1 + g)u'(f(l_{t+1}))f'(l_{t+1}).$$

A steady state with constant extraction $l_t = l^* > 0$ requires

$$\beta(1 + g) = 1.$$

For the log/Cobb-Douglas case, $u(c) = \log c$ and $f(l) = l^\gamma$, so current utility is $\gamma \log l$. Guess

$$v(x) = a + b \log x.$$

FOC:

$$\frac{\gamma}{l} = \frac{\beta b}{x - l}.$$

Coefficient matching gives

$$b = \gamma + \beta b \implies b = \frac{\gamma}{1 - \beta}.$$

Thus

$$l = (1 - \beta)x, \quad x' = \beta(1 + g)x, \quad c = l^\gamma = [(1 - \beta)x]^\gamma.$$

Consumption is constant over time if and only if

$$\beta(1 + g) = 1.$$

13.6 Human capital accumulation

The problem is

$$v(s) = \max_{0 \leq i \leq ws} \{u(ws - i) + \beta v((1 - \delta)s + As^{1-\gamma}i^\gamma)\}.$$

The state is skill s ; the control is investment i ; consumption is $c = ws - i$.

For general u , the FOC is

$$u'(c) = \beta v'(s') A \gamma s^{1-\gamma} i^{\gamma-1},$$

and the envelope condition is

$$v'(s) = u'(c)w + \beta v'(s') [(1 - \delta) + A(1 - \gamma)s^{-\gamma}i^\gamma].$$

For $u(c) = \log c$, guess

$$v(s) = a + B \log s, \quad i = js.$$

Then $c = s(w - j)$ and

$$s' = s[(1 - \delta) + Aj^\gamma].$$

Substitute into the Bellman equation:

$$a + B \log s = \max_j \{\log s + \log(w - j) + \beta a + \beta B \log s + \beta B \log((1 - \delta) + Aj^\gamma)\}.$$

Coefficient matching gives

$$B = 1 + \beta B = \frac{1}{1 - \beta}.$$

The scalar FOC for j is

$$\frac{1}{w - j} = \frac{\beta B A \gamma j^{\gamma-1}}{(1 - \delta) + Aj^\gamma}.$$

Using $B = 1/(1 - \beta)$,

$$(1 - \beta) [(1 - \delta) + Aj^\gamma] = \beta A \gamma j^{\gamma-1} (w - j).$$

Thus optimal investment is linear in skill:

$$i^*(s) = j^* s.$$

Because the objective in j is strictly concave, the FOC has a unique solution. If $F(j, \delta) = 0$ denotes the FOC rearranged as

$$F(j, \delta) = \beta A \gamma j^{\gamma-1} (w - j) - (1 - \beta) [(1 - \delta) + Aj^\gamma],$$

then $F_\delta = 1 - \beta > 0$ and the second-order condition gives $F_j < 0$ at the optimum. Hence

$$\frac{\partial j^*}{\partial \delta} = -\frac{F_\delta}{F_j} > 0.$$

Higher depreciation raises replacement investment as a share of skill.

13.7 Survival-risk project choice

There are two projects $i \in \{S, R\}$. If the project survives, investment k' pays $A_i k'$ next period with probability γ_i . If it fails, the agent receives benefit b forever. Let

$$B_0 = \frac{\log b}{1 - \beta}$$

be the failure value. Conditional on committing to project i , the Bellman equation is

$$V_i(k) = \max_{0 \leq k' \leq k} \left\{ \log(k - k') + \beta [\gamma_i V_i(A_i k') + (1 - \gamma_i) B_0] \right\}.$$

Guess

$$V_i(k) = a_i + B_i \log k.$$

The FOC is

$$\frac{1}{k - k'} = \frac{\beta \gamma_i B_i}{k'}.$$

The coefficient condition is

$$B_i = 1 + \beta \gamma_i B_i \implies B_i = \frac{1}{1 - \beta \gamma_i}.$$

Therefore

$$k'_i = \beta \gamma_i k, \quad c_i = (1 - \beta \gamma_i)k.$$

The value difference is

$$V_S(k) - V_R(k) = (a_S - a_R) + (B_S - B_R) \log k.$$

Since $\gamma_S > \gamma_R$, $B_S > B_R$, so the value difference is increasing in $\log k$. Therefore there is a cutoff $\bar{k}$ such that the agent chooses the safer project S if and only if $k \geq \bar{k}$, allowing for corner cases where one project dominates for all k .

Qualifier trap

Even if $A_S > A_R$ and $\gamma_S > \gamma_R$, initial consumption under S need not be higher. Conditional saving is $k'_i = \beta \gamma_i k$, so higher survival probability raises saving and lowers current consumption share: $c_i = (1 - \beta \gamma_i)k$.

13.8 Disaster-risk AK with CRRA utility

Let $\eta \in \{1, \rho\}$ be the surviving fraction of capital after the beginning-of-period disaster, with

$$\Pr(\eta = \rho) = q, \quad \Pr(\eta = 1) = 1 - q.$$

Effective capital is ηk , output is $A\eta k$, and the resource constraint is

$$c + k' = A\eta k.$$

The Bellman equation can be written as

$$V(k) = \mathbb{E}_\eta \left[\max_{0 \leq k' \leq A\eta k} \{u(A\eta k - k') + \beta V(k')\} \right].$$

For CRRA utility,

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0, \quad \sigma \neq 1,$$

try a constant saving share from realized output:

$$k' = sA\eta k, \quad c = (1-s)A\eta k.$$

The Euler equation is

$$u'(c_t) = \beta \mathbb{E}_t [u'(c_{t+1}) A\eta_{t+1}].$$

Using $u'(c) = c^{-\sigma}$ and the policy above,

$$1 = \beta s^{-\sigma} A^{1-\sigma} \mathbb{E} [\eta^{1-\sigma}].$$

Therefore

$$s = [\beta A^{1-\sigma} ((1-q) + q\rho^{1-\sigma})]^{1/\sigma}.$$

An interior solution requires $0 < s < 1$.

Comparative statics:

- If $\sigma > 1$, then $\rho^{1-\sigma} > 1$, so higher q raises s and a lower ρ raises s .
- If $0 < \sigma < 1$, then $\rho^{1-\sigma} < 1$, so higher q lowers s and a lower ρ lowers s .
- The log case $\sigma = 1$ is the limiting case with $s = \beta$, independent of q and ρ .

13.9 Alternating management rights

With independent management and log utility, each agent solves

$$V(k) = \max_{0 \leq k' \leq Ak} \{\log(Ak - k') + \beta V(k')\},$$

so

$$k' = \beta Ak.$$

Aggregating across the two independently managed halves gives

$$K' = \beta AK.$$

With joint ownership and alternating management, agent i solves

$$V_i(k) = \max_{c, c', s, k' \geq 0} \{ \log c + \beta \log c' + \beta^2 V_i(g_{-i}(k')) \}$$

subject to

$$c + s + k' = Ak, \quad c' = As.$$

Look for a symmetric linear rule $g_i(k) = g(k) = \eta Ak$ and value $V(k) = a + B \log k$. The objective becomes, up to constants,

$$\log c + \beta \log s + \beta^2 B \log k'.$$

The optimal shares of Ak are proportional to the log weights 1, β , and $\beta^2 B$:

$$\frac{c}{Ak} = \frac{1}{1 + \beta + \beta^2 B}, \quad \frac{s}{Ak} = \frac{\beta}{1 + \beta + \beta^2 B}, \quad \frac{k'}{Ak} = \frac{\beta^2 B}{1 + \beta + \beta^2 B}.$$

Coefficient matching gives

$$B = 1 + \beta + \beta^2 B \implies B = \frac{1 + \beta}{1 - \beta^2} = \frac{1}{1 - \beta}.$$

Hence

$$\frac{k'}{Ak} = \beta^2, \quad K' = \beta^2 AK.$$

Since $\beta^2 < \beta$, the joint alternating-rights arrangement grows more slowly than independent management.

14 Closed-form catalog

Model	Bellman	Policy / equation	Verification point
Quadratic reaching-center	$v(x) = \min_y \{ (y-x)^2 + y^2 + \beta v(y) \}$	$y = x/(2+\beta A), A = (1+\beta A)/(2+\beta A)$	Positive root unique.
Cake-eating	$v(x) = \max_c \{ \log c + \beta v(x-c) \}$	$c = (1-\beta)x$	$b = 1/(1-\beta)$.
Cobb-Douglas growth	$v(k) = \max_{k'} \{ \log(Ak^\alpha k') + \beta v(k') \}$	$= k' = \alpha \beta A k^\alpha$ —	$b = \alpha/(1-\alpha\beta)$.
AK log growth	$v(k) = \max_{k'} \{ \log(Ak k') + \beta v(k') \}$	$= k' = \beta A k$ —	Requires βA consistent with desired path/transversality.
Stochastic growth	$v(k, z) = \max_{k'} \{ U(zk^\alpha k') + \beta E v(k', z') \}$	$=$ Euler with expectation over z' —	Need compact invariant K .

Model	Bellman	Policy / equation	Verification point
Growth with leisure	$F(k, k')$ $\max_n U(f(k, n)$ $(1 - \delta)k - k', 1 - n)$	$= U_c f_n = U_\ell$ $+$	Do not put labor as next state.
Tree-cutting	$v(x)$ $\max\{x, \beta v(h(x))\}$	$=$ Cut when $x \geq \beta v(h(x))$	Prove cutoff by value comparison.
Resource extraction	$v(x) = \max_l \{\gamma \log l +$ $\beta v((x - l)(1 + g))\}$	$l = (1 - \beta)x$	Constant consumption iff $\beta(1 + g) = 1$.
Human capital	$v(s) = \max_i \{\log(ws -$ $i) + \beta v(s')\}$	$i = j^* s$	j^* solves scalar FOC.
Survival project	$V_i(k) = \max_{k'} \{\log(k -$ $k') + \beta[\gamma_i V_i(A_i k') + (1 - \gamma_i)B_0]\}$	$k' = \beta \gamma_i k$	Cutoff because $B_S > B_R$.
Disaster AK	$u'(c) = \beta E[u'(c')A\eta']$	$s = [\beta A^{1-\sigma} E\eta^{1-\sigma}]^{1/\sigma}$	Comparative statics depend on σ .
Alternating management	$V_i(k)$ $\max\{\log c + \beta \log c' +$ $\beta^2 V_i(g_{-i}(k'))\}$	$= K' = \beta^2 AK$	Strategic delay lowers saving.

15 Core derivation templates

15.1 Template A: static KKT

1. Write the objective and constraints with signs chosen consistently.
2. Write $\mathcal{L} = \text{objective} + \sum \mu_i \times \text{slack}_i$.
3. State stationarity, feasibility, nonnegativity, and complementary slackness.
4. State concavity/convexity conditions for sufficiency.
5. Interpret the multiplier.

15.2 Template B: recursive formulation

1. State timing: when shocks are observed, when controls are chosen, when payoffs occur.
2. State variables: all payoff-relevant predetermined variables.
3. Controls: all current choices.
4. Feasible correspondence $\Gamma(x)$.
5. Bellman equation.
6. Policy functions and transition laws.

15.3 Template C: FOC-envelope-Euler

$$\begin{aligned}
 v(k) &= \max_{k'} \{u(F(k) - k') + \beta E v(k', z')\}, \\
 u'(c) &= \beta E[v_k(k', z')], \\
 v_k(k, z) &= u'(c) F_k(k, z), \\
 u'(c_t) &= \beta E_t[u'(c_{t+1}) F_k(k_{t+1}, z_{t+1})].
 \end{aligned}$$

15.4 Template D: coefficient matching

1. Guess $v(x) = a + b \log x$, $v(x) = Ax^2$, or $V(k) = Dk^{1-\sigma} + C$.
2. Substitute into Bellman.
3. Solve FOC for policy in terms of unknown coefficients.
4. Substitute policy back.
5. Match coefficient on the state term and constant term.
6. Verify feasibility and uniqueness.

16 Exam variations

16.1 What changes and what does not

Variation	What changes	What stays invariant
Leisure or labor choice	Add intratemporal control and time constraint	State is still capital unless labor has persistence.
Stochastic productivity	Add shock to state and expectation over next shock	Same FOC-envelope-Euler structure.
Full depreciation	Resource constraint becomes $c + k' = zk^\alpha$	Euler uses next marginal product only.
Stopping	Control is discrete: stop or continue	Bellman is max of stop value and continuation value.
Project survival	Add failure state and survival probability	Conditional value often log-linear; compare project values.
Irreversible choice	Initial discrete choice affects all future values	Solve conditional values, then compare at $t = 0$.
Externality/public leisure	Planner internalizes object that private agents may not	FOCs reveal wedge; recursive setup unchanged.
Strategic management	Other player's policy enters continuation value	Guess symmetric policy, then verify fixed point.

16.2 Past-qual practice map

Year	Problem	Module 1 skill	How to practice it
2016	Q1	Planner growth with multiple households	Write planner problem, Pareto weights, aggregate capital law, and show aggregate path depends on K_0 not distribution.
2016	Q4	Growth/Ramsey with taxes	Derive household FOCs and implementability; compare CE and planner conditions.
2017	Q2	Endogenous growth with externality	Derive competitive and efficient growth rates; identify policy correction.
2017	Q4	Static labor tax smoothing	Practice static FOCs, government budget, and welfare comparison.
2018	Q3	Natural resource extraction	Derive sequential problem, Euler equation, log closed form, steady-state condition.
2019	Q3	Work/non-work irreversible choice with saving	Set up dynamic problem, borrowing constraint, policy comparison, cutoff in assets.
2020	Q1	Growth with public leisure	Recursive planner, FOC, envelope, steady-state capital-labor ratio, log hours.
2021	Q3	Skill accumulation	Recursive setup, Euler, log guess-and-verify linear investment, δ comparative static.
2022	Q1	Project survival choice	Recursive setup, log value functions, cutoff proof, initial consumption comparison.
2023	Q3	AK management rights	Recursive setup, independent log AK saving, symmetric linear equilibrium under alternating control.
2024	Q1	Disaster-risk AK	Stochastic recursive planner, Euler equation, CRRA saving share, disaster comparative statics.

17 Common traps

1. **Wrong state:** choosing consumption as a state when it is a control, or omitting a persistent shock.
2. **Wrong timing:** using $f'(k_t)$ in the Euler equation when the return on saving is $f'(k_{t+1}) + 1 - \delta$.
3. **No feasible correspondence:** writing a Bellman equation without $k' \in \Gamma(k)$.

4. **FOC without sufficiency:** deriving conditions but never invoking concavity.
5. **Envelope with policy derivatives:** differentiating $k'(k)$ inside the envelope condition.
6. **Unbounded contraction:** claiming contraction on an unbounded space without bounded returns or a compact invariant state.
7. **Guess not verified:** stating $v(k) = a + b \log k$ without matching coefficients or checking feasibility.
8. **Ignoring KKT:** assuming interiority when nonnegativity or borrowing constraints are central.
9. **Cutoff asserted, not proved:** a cutoff requires a monotone value difference.
10. **Pure intuition on Economist X:** the grader wants a formal sign or comparison first.

18 Exercise bank

18.1 Mechanical drills

- M1. Skill: KKT.** Solve $\max_{c,k'} \log c + \beta V(k')$ subject to $c + k' \leq y$, $c \geq 0$, $k' \geq 0$. Write the KKT conditions and interpret the multiplier on $k' \geq 0$.
- M2. Skill: identify states.** For each variable, label it state, control, shock, or parameter in: stochastic growth with Markov z , human capital $s' = (1 - \delta)s + As^{1-\gamma}i^\gamma$, and tree-cutting $x' = h(x)$.
- M3. Skill: abstract DP mapping.** Map growth with leisure into $v(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}$.
- M4. Skill: envelope.** For $v(k) = \max_{k'} u(Ak^\alpha - k') + \beta v(k')$, derive the envelope condition.
- M5. Skill: stochastic Euler.** Derive the Euler equation for $v(k, z) = \max_{k'} \{u(zk^\alpha - k') + \beta E[v(k', z') | z]\}$ when z is Markov.
- M6. Skill: bounded state.** Find an invariant interval $[0, \bar{k}]$ for $k' = zk^\alpha - c$, $z \leq \bar{z}$, full depreciation.
- M7. Skill: monotonicity.** Explain why the value function in deterministic growth is increasing in k .
- M8. Skill: single-valued policy.** State sufficient conditions that make $G(k)$ single-valued in the deterministic growth problem.

18.2 Closed-form and proof drills

- C1. Skill: quadratic guess.** Re-derive the infinite-horizon reaching-the-center coefficient A and policy x' .
- C2. Skill: log cake-eating.** Solve the cake-eating problem and show $c_t = (1 - \beta)x_t$.
- C3. Skill: log Cobb-Douglas.** Solve $v(k) = \max_{k'} \{\log(Ak^\alpha - k') + \beta v(k')\}$.
- C4. Skill: resource extraction.** Solve the log resource problem with $f(l) = l^\gamma$ and identify when consumption is constant.
- C5. Skill: human capital.** Derive the scalar equation for j in the policy $i = js$.

- C6. Skill: project survival.** Conditional on project i , show $k'_i = \beta\gamma_i k$.
- C7. Skill: disaster CRRA.** Derive the saving share s and sign its derivative with respect to q for $\sigma > 1$ and $\sigma < 1$.
- C8. Skill: contraction.** Prove Blackwell discounting for a Bellman operator.
- C9. Skill: concavity.** Prove that T preserves concavity in the deterministic growth problem.
- C10. Skill: cutoff proof.** Suppose $D(k) = a + b \log k$ with $b > 0$. Prove a cutoff rule for choosing option S over R .

18.3 Past-qual-style applications

- P1. Skill: 2018-style natural resource.** Set up the sequential problem, derive the Euler equation, and solve the log/Cobb-Douglas case.
- P2. Skill: 2020-style growth with public leisure.** Write the recursive planner problem when utility is $u(c) + \phi(\ell, \bar{\ell})$ and $\ell = \bar{\ell}$ under the planner.
- P3. Skill: 2021-style human capital.** Use guess-and-verify to prove linear investment in skill.
- P4. Skill: 2022-style project cutoff.** Show safer project choice is characterized by a threshold in initial capital.
- P5. Skill: 2023-style alternating management.** Derive $K' = \beta^2 AK$ under symmetric linear policies.
- P6. Skill: 2024-style disaster.** Write the recursive problem when the disaster shock is realized before production and derive the Euler equation.

18.4 Trap and Economist X questions

- T1. Skill: state-variable trap.** Economist X says a stochastic growth problem with iid z can use k as the only state because z has no persistence. Is this correct if z is observed before the saving choice?
- T2. Skill: sufficiency trap.** A classmate derives an FOC and declares the optimum unique. What additional conditions are needed?
- T3. Skill: policy intuition trap.** In the survival-project model, a classmate says the safer project must have higher initial consumption because it is better. Is this correct?
- T4. Skill: disaster intuition.** Economist X says higher disaster probability always raises saving. Is this true in the CRRA disaster AK model?
- T5. Skill: stopping trap.** A student says the tree should be cut when $x \geq \beta h(x)$. Is this always the Bellman stopping condition?
- T6. Skill: envelope trap.** A student writes $v'(k) = u'(c)(f'(k) - g'(k)) + \beta v'(g(k))g'(k)$. Explain the error.

19 Answer sketches

19.1 Mechanical drills

- M1.** Lagrangian: $\mathcal{L} = \log c + \beta V(k') + \lambda(y - c - k') + \mu_c c + \mu_k k'$. Interior c gives $1/c = \lambda$. If $k' > 0$, $\beta V'(k') = \lambda$; if $k' = 0$, then $\beta V'(0) \leq \lambda$. A positive μ_k means the nonnegativity constraint prevents lower saving/borrowing.
- M2.** Stochastic growth: k, z are states if z observed before choice, k' is control, z' future shock. Human capital: s state, i control, w, A, γ, δ parameters. Tree: x state, cut/wait control.
- M3.** Use $x = k$, $y = k'$, and $F(k, k') = \max_{n \in [0,1]} U(f(k, n) + (1 - \delta)k - k', 1 - n)$ with feasible k' .
- M4.** FOC is $u'(c) = \beta v'(k')$. Envelope is $v'(k) = u'(c)\alpha A k^{\alpha-1}$.
- M5.** FOC: $u'(c) = \beta E[v_k(k', z') \mid z]$. Envelope next period: $v_k(k', z') = u'(c')\alpha z'(k')^{\alpha-1}$. Combine.
- M6.** Choose $\bar{k} \geq \max\{k_0, 1, \bar{z}^{1/(1-\alpha)}\}$ so $zk^\alpha \leq \bar{k}$ for $k \leq \bar{k}$.
- M7.** More k expands feasible consumption and saving; using the old policy is feasible and gives at least as much utility, strictly more if extra resources can raise consumption.
- M8.** Strict concavity of u , concavity of f , convex feasible set, and strict concavity of the maximand in k' imply a unique policy; continuity follows by the theorem of the maximum.

19.2 Closed-form and proof drills

- C1.** Guess $v = Ax^2$. FOC gives $x' = x/(2 + \beta A)$. Coefficient equation $A = (1 + \beta A)/(2 + \beta A)$ gives positive root $A = (\beta - 2 + \sqrt{\beta^2 + 4})/(2\beta)$.
- C2.** Guess $a + b \log x$. Coefficient $b = 1/(1 - \beta)$; FOC gives $c = x/(1 + \beta b) = (1 - \beta)x$.
- C3.** Guess $a + b \log k$. Match $b = \alpha(1 + \beta b)$, so $b = \alpha/(1 - \alpha\beta)$. Policy $k' = \alpha\beta A k^\alpha$.
- C4.** Bellman $v(x) = \max_l \{\gamma \log l + \beta v((x - l)(1 + g))\}$. Guess $a + b \log x$, $b = \gamma/(1 - \beta)$, $l = (1 - \beta)x$, and $x' = \beta(1 + g)x$. Consumption is constant iff $\beta(1 + g) = 1$.
- C5.** Substitute $i = js$ and $v = a + B \log s$. $B = 1/(1 - \beta)$ and j solves $(1 - \beta)[(1 - \delta) + Aj^\gamma] = \beta A \gamma j^{\gamma-1}(w - j)$.
- C6.** Guess $V_i = a_i + B_i \log k$. $B_i = 1/(1 - \beta\gamma_i)$ and FOC gives $k'_i = \beta\gamma_i k$.
- C7.** $s = [\beta A^{1-\sigma}((1 - q) + q\rho^{1-\sigma})]^{1/\sigma}$. If $\sigma > 1$, $\rho^{1-\sigma} > 1$, so s rises with q . If $\sigma < 1$, it falls.
- C8.** $T(v + a) = \max_y \{F + \beta(v(y) + a)\} = Tv + \beta a$. Together with monotonicity, this implies contraction.
- C9.** For $\theta \in [0, 1]$, convex combinations of feasible plans are feasible; concavity of u and f gives $Tv(\theta k_1 + (1 - \theta)k_2) \geq \theta Tv(k_1) + (1 - \theta)Tv(k_2)$.
- C10.** $D(k) = a + b \log k$ is increasing in k . If $D(\bar{k}) = 0$, then choose S for $k \geq \bar{k}$ and R for $k < \bar{k}$; corner cases if D never crosses zero.

19.3 Past-qual-style applications

- P1.** Use state x and control l . Euler: $u'(f(l_t))f'(l_t) = \beta(1+g)u'(f(l_{t+1}))f'(l_{t+1})$. In the log case, $l = (1-\beta)x$.
- P2.** State K or per-capita k . Planner internalizes $\bar{\ell} = \ell$. Bellman includes $u(c) + \phi(\ell, \ell) + \beta v(K')$ subject to aggregate feasibility and $h + \ell = 1$.
- P3.** Same as C5. The key is to scale by s : $c = s(w - j)$ and $s' = s[(1-\delta) + Aj^\gamma]$, which makes the log guess valid.
- P4.** Same as C6 and C10. Since $V_S - V_R = (a_S - a_R) + (B_S - B_R) \log k$ with $B_S > B_R$, the difference is increasing.
- P5.** Guess $V = a + B \log k$. Shares of Ak are proportional to $1, \beta, \beta^2 B$. Matching gives $B = 1/(1-\beta)$ and $k' = \beta^2 Ak$.
- P6.** Let $\eta \in \{1, \rho\}$. Bellman $V(k) = E_\eta \max_{k'} \{u(A\eta k - k') + \beta V(k')\}$. Euler $u'(c) = \beta E[u'(c')A\eta]$.

19.4 Trap and Economist X questions

- T1.** Not correct. Even iid z affects current feasible resources if observed before choice, so the current state is (k, z) . Iid only means the conditional distribution of z' does not depend on current z .
- T2.** Need concavity/convex feasible set for sufficiency and strict concavity for uniqueness, or a direct global argument.
- T3.** Not necessarily. Conditional on project i , $c_i = (1 - \beta\gamma_i)k$. The safer project has higher γ_i and therefore lower current consumption share.
- T4.** False. In the CRRA disaster AK model, the sign depends on σ . Higher disaster probability raises saving only when $\sigma > 1$.
- T5.** Not always. The Bellman condition is $x \geq \beta v(h(x))$, not $x \geq \beta h(x)$ unless the next state is surely in the stopping region.
- T6.** The envelope theorem omits policy derivatives because the control is optimized. The correct envelope is the direct derivative of the maximand with respect to the state.

20 Final study checklist

Before attempting old quals, confirm that you can do each item without notes:

- ☐ Write a KKT system with correct multiplier signs and complementary slackness.
- ☐ Map a sequential problem into a Bellman equation with state, control, feasible correspondence, and transition.
- ☐ Explain why payoff separability and Markov transitions justify recursive formulation.
- ☐ Derive FOC, envelope, and Euler equations for deterministic growth.
- ☐ Derive the stochastic Euler equation for stochastic growth.
- ☐ Construct an invariant bounded state space for full-depreciation stochastic growth.

- ☐ Prove a Bellman operator is a contraction using Blackwell conditions.
- ☐ Prove value monotonicity and concavity using the Bellman operator.
- ☐ Prove policy single-valuedness from strict concavity and continuity from the theorem of the maximum.
- ☐ Solve the quadratic reaching-the-center problem by coefficient matching.
- ☐ Solve log cake-eating, log Cobb-Douglas growth, and log AK growth from scratch.
- ☐ Solve the natural resource extraction log case and state the steady-state condition.
- ☐ Derive the human-capital linear investment rule under log utility.
- ☐ Derive the project-survival saving rule and cutoff logic.
- ☐ Derive the disaster-risk AK saving share and sign comparative statics.
- ☐ Derive alternating-management growth $K' = \beta^2 AK$.
- ☐ Answer an Economist X claim with an equation first and intuition second.

Quality audit

- Source inventory: included, with caveat that separate Galina lecture notes/slides were not accessible.
- Module map: included.
- Self-contained canonical derivations: included for static optimization, deterministic DP, stochastic DP, stopping, and closed-form labs.
- Current PS1 emphasis: reaching-the-center, tree-cutting, growth with leisure, stochastic growth, bounded state restrictions, contraction, concavity, monotonicity, and single-valued policies are explicitly covered.
- Past-qual mappings: included by year and problem.
- Closed forms: derived through guess-and-verify rather than merely listed.
- Exercises: categorized and paired with answer sketches.
- Common traps: explicit.
- TeX: uses standard Overleaf packages, compact lists, booktabs/longtable, and display math only for important equations.